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Review

The impact of aerobic exercise on depression in young people: A meta regression and meta-analysis☆

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ABSTRACT

Objective: This systematic review aimed to 1) update the existing evidence on the antidepressant effects of aerobic exercise (AE) in youth. 2) Investigate any potential dose–response relationships between AE interventions and depressive symptom reduction. 3) Provide evidence-based insights to inform future research and clinical depression treatment.

Methods: Employing the PRISMA and PERSiST guidelines, a comprehensive search across nine databases (Web of Science, PubMed, Scopus, PsycINFO, SportDiscus, CINAHL, Medline, Embase, and CNKI) yielded a total of 782 relevant studies. Following rigorous selection criteria, 26 eligible studies (comprising 22 different samples) were included in the analysis, featuring a combined sample size of 1308 participants. The meta-analysis was conducted via R.

Results: AE notably decreased depressive symptoms among youth [$g = -0.92$; 95% CI (−1.16, −0.69); $p < 0.01$]. Significant dose–response relationships were observed across age groups [$\beta = -0.06$; 95% CI (−0.12, 0.00), $p < 0.05$], the intensity [$\beta = -0.03$; 95% CI (−0.06, 0.00); $p < 0.05$], and the duration ˆ 2 [$\beta = 0.001$, 95% CI (0.001, 0.002), $p < 0.05$] of the AE intervention.

Conclusions: Aerobic exercise interventions can effectively alleviate depressive symptoms in youth. A moderate-intensity AE program, lasting 25–40 min and conducted three times a week for 9–15 weeks, can increase alleviation of depression.

1. Introduction

Depression is a prevalent mental health disorder affecting millions of people globally and is characterized by persistent sadness, hopelessness, and disinterest in previously enjoyable activities (Guerrera et al., 2023). Depression can also lead to debilitating outcomes, including diminished quality of life, comorbid medical conditions, and increased mortality risk (Wise, 2019). As reported in 2017, approximately 4.4% of the global population—or more than 300 million individuals—were living with depression (World Health Organization, 2017). Epidemiological studies have revealed that approximately 20% of young people worldwide experience mental health issues, underscoring the importance of addressing depression within this age group (Lu, 2019).

Studies have shown that depression escalates most rapidly among adolescents and young adults, with a notable spike during adolescence (Goodwin et al., 2022). Specifically, the prevalence of depression increases from 0.4 to 2.5% in childhood to 8.3% in adolescence (Birmaher et al., 1996). Concomitantly, subclinical depressive symptoms, which do not meet the full diagnostic criteria for depression, also increase significantly among adolescents. The coronavirus disease (COVID-19) pandemic has led to significant disruptions in adolescents' daily lives worldwide, worsening existing stressors (Lee, 2020). In particular, adolescents have faced prolonged social isolation, missed milestones, school closures, and increased family stress, which are all known contributors to psychological distress and mental health challenges in this age group (Brooks et al., 2020; Golberstein et al., 2020; Loades et al., 2020). Untreated depression has severe consequences, including social dysfunction, health decline, and increased risk of suicide (Klein et al.,

Abbreviations: AE, Aerobic exercise; BDI, Beck Depression Inventory; PHQ-9, Patient Health Questionnaire-9; SDS, Self-rating Depression Scale; Zung, Zung self-rating depression; CES-D, scale Center for Epidemiologic Studies Depression Scale; HAM-D, Hamilton Rating Scale for Depression; MDD, Major depressive disorder; HRmax, Maximum heart rate; HRR, Heart rate reserve; BDNF, Brain derived neurotrophic factor.

☆ This study was registered in the International Prospective Register of Systematic Reviews (PROSPERO) with the protocol number CRD42024500427.

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2009; Windfuhr et al., 2008). Consequently, it is a leading contributor to the global disease burden among young people under 25 years of age (Gore et al., 2011). If left unchecked, depression can deteriorate, leading to the development of comorbid health issues and escalating societal costs (Kessler, 2012). Hence, it is essential to prioritize treatments that are effective, accessible, and cost-efficient (Bryant et al., 2021).

According to current clinical guidelines, evidence-based treatments for depression among young individuals include psychotherapy and pharmacotherapy (Avenevoli et al., 2015; Kataoka et al., 2002; Malhi et al., 2021). One such psychotherapy is cognitive behavioral therapy (CBT), an approach that helps individuals recognize, explore, and modify the relationships among negative thoughts, behaviors, and depressive emotions. Although CBT has been found to be effective in changing negative thought patterns, its overall effectiveness in real-world settings is somewhat limited (Hetrick et al., 2016). The same phenomenon is observed with the antidepressant fluoxetine, which is used as a pharmacotherapy (Cipriani et al., 2016; Hetrick et al., 2012; Pasquini et al., 2009). Furthermore, psychotherapy often comes with an excessive cost and only achieves a remission rate of approximately 50%, whereas antidepressants can lead to side effects, a risk of relapse, and withdrawal symptoms (Andrews et al., 2000; March et al., 2005; Schuch et al., 2016). Other barriers to accessing psychotherapy for youth may be related to limited treatment options, concerns about stigma, parental reservations about therapy, and misconceptions about the treatment process itself (Kvam et al., 2016). Moreover, antidepressants can trigger adverse effects in young patients, including sleep disturbances, gastrointestinal issues, and even suicidal tendencies (Milev et al., 2016). Considering these challenges, there is an urgent need to explore innovative treatment strategies that can be easily incorporated into the daily routines of young people with depression.

Lifestyle medicine is emerging as a promising alternative, offering convenient and readily accepted therapies. Among these interventions, physical activity interventions have gained traction in the management of mental health disorders and have shown efficacy in reducing depressive symptoms (Kessler, 2012; Sarris et al., 2014). The WHO and the National Institute for Health and Care Excellence (NICE) in the UK endorse exercise as a complementary therapy for depression (Birmaher & Brent, 2007; Bull et al., 2020). Compared with traditional psychological and pharmacological treatments, physical activity is more accessible and acceptable and can be better controlled throughout the 24 h of the day (Zhang & Li, 2023).

Systematic reviews focusing on children and adolescents revealed that physical activity has varying degrees of antidepressant effects, ranging from minimal to significant (Bailey et al., 2018; Recchia et al., 2023; Zhang & Li, 2023). However, disparities across studies are notable and attributed to methodological and conceptual variances, including differences in inclusion criteria and analytical approaches. Notably, certain investigations have focused exclusively on individuals diagnosed with major depressive disorder (MDD) (Bailey et al., 2018), which might limit the generalizability of the findings to other populations experiencing depressive symptoms. The focus on MDD might also overlook the potential benefits of physical activity for youth who do not meet diagnostic criteria. In addition, combined exercise (e.g., aerobic exercise combined with resistance exercise) in the intervention group may distort the overall effect of a given type of physical activity (Zhang et al., 2023). An umbrella review recently highlighted the promising potential of aerobic exercise (AE) in treating adolescent depression. Moreover, the precise mechanisms, regulatory roles, and dose–response effects of AE prescriptions require further investigation (Wegner et al., 2020). Hence, the aims of this study were to 1) update the existing body of evidence on the antidepressant effects of AE in youth. 2) Investigate any potential dose–response relationships between AE interventions and depressive symptom reduction. 3) Provide evidence-based insights to inform future research and clinical depression treatment.

2. Method

This systematic review and meta-analysis were conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines (Page et al., 2021), while also incorporating the PERSiST guidance, which aims to optimize the application of PRISMA in the fields of exercise, rehabilitation, sports medicine, and exercise science (Ardern et al., 2022). This systematic review and meta-analysis were registered in the PROSPERO database (CRD42024500427).

2.1. Search strategy

The literature search and selection adhered to the PRISMA criteria and guidelines and involved a thorough investigation across several databases: Web of Science, PubMed, Scopus, PsycINFO, SportDiscus, CINAHL, Medline, Embase, and CNKI. The search spanned from database establishment to January 28, 2024. Keywords employed included combinations of terms related to adolescence and physical activity, such as Adolescent OR Adolescence OR Teenager OR Youth OR juvenile OR student OR undergraduate OR “college student” OR “university student” AND “physical activity” OR “physical exercise” OR sport OR running OR walking OR jogging OR cycling OR swimming OR “aerobic exercise” OR “aerobic training” AND “depression OR “depressive disorder” OR “depressive syndrome” OR “Depressive Syndromes” OR melancholia. References from the retrieved articles were also examined to augment the search.

2.2. Study selection

The eligibility of the studies for inclusion was independently assessed by two authors (QY and YL), and the final decision regarding inclusion was made after a summary of the comments was prepared and discussed with another author (JW). The reference lists of all included studies were examined during the secondary search.

The literature screening was based on the PICOS criteria for evidence-based medicine, and the specific literature inclusion criteria were as follows: (1) participants were younger than 24 years, and no restrictions on the clinical diagnosis of depression were made at baseline to ensure accuracy in assessing the effect of AE on depression prevention. (2) The intervention group was assigned to AE, which is defined as physical activity under conditions with adequate oxygen supply (Mersy, 1991), with no restrictions on exercise intensity. (3) The control group included nonexercise controls, such as no intervention, usual care, wait-list controls, or stretching. (4) The scales used to measure depression included the Beck Depression Inventory (BDI), Patient Health Questionnaire-9 (PHQ-9), Self-rating Depression Scale (SDS), Zung self-rating depression scale (Zung), Center for Epidemiologic Studies Depression Scale (CES-D), and Hamilton Rating Scale for Depression (HAM-D). (5) The type of experiment should be a before-and-after controlled trial. The excluded studies were nonoriginal studies such as reviews and meta-analyses, conference abstracts, dissertations, and clinical trials that did not report enough data for meta-analysis.

2.3. Data extraction

The initial extraction of relevant data was conducted by two authors (QY and YL), with a third author serving as an additional check (JW). A systematic data extraction form was employed to record pertinent information from each study, including (1) the first author and year of publication; (2) sample characteristics (e.g., sample size, mean age of participants, baseline clinical diagnoses of depression, and chronic diseases); (3) details of the interventions (e.g., type of AE, duration and frequency of intervention, intensity and duration of AE); (4) methodological considerations (e.g., risk of bias, diagnostic tools, and symptom assessment methods); and (5) primary outcomes (e.g., changes in total depressive symptom scores preintervention and postintervention).

The intensity of the AE was collected on the basis of the maximum heart rate (HRmax) and heart rate reserve (HRR). The samples were classified as light-intensity (≤39% HRR or 63% HRmax), moderate-intensity (40–59% HRR or 64–76% HRmax), or vigorous-intensity (≥60% HRR or 77% HRmax) according to the criteria established by the American College of Sports Medicine (Garber et al., 2011). We categorized the age of the population into childhood (<10 years), early adolescence (10–14 years), late adolescence (15–19 years), and early adulthood (19–24 years) with reference to previous studies (van Lang et al., 2007). Additionally, we categorized the participants according to their stage of education, such as elementary school, high school, or college.

2.4. Methodological quality of included studies

Two authors (QY and YL) used the 11-item Physiotherapy Evidence Database (PEDro scale) for independent assessment, and discrepancies were addressed by consulting a third author (JW). The PEDro scale includes 11 questions, where each question beyond the first is awarded one point for compliance with its criteria, leading to a maximum achievable score of 10 points. Scores are allocated on the basis of adherence to these criteria: 1 point for compliance and 0 point for noncompliance. The aggregated scores categorize the literature's quality as excellent (9–10 points), good (6–8 points), fair (4–5 points), or poor (≤3 points) (Maher et al., 2003).

2.5. Statistical analysis

For descriptive statistics, we used the standardized mean difference (with the standard deviation, SD) as a measure of central tendency. Hedges' *g* method was used to estimate the standardized mean difference (SMD). The sizes of Hedges' *g* effects can be categorized into small effects ($g = 0.2$), medium effects ($g = 0.5$), and large effects ($g = 0.8$). Given the heterogeneity caused by study differences (e.g., demographics or training protocol), a random effects model was employed for the meta-analysis. This approach was selected over the fixed-effects model because of its capacity to account for the variability of true effect sizes and its suitability for included studies (Dettori et al., 2022).

We used Cochran's Q statistic (Q-test) to determine heterogeneity; additionally, τ^2 was calculated to quantify the variance between studies, and I^2 was employed to evaluate the percentage of variance attributed to between-study heterogeneity. The size of I^2 was classified as low ($I^2 < 25\%$), moderate ($25\% < I^2 < 75\%$), or high ($I^2 > 75\%$). Forest plots were designed to report the results of the meta-analysis.

Sensitivity analyses were developed to detect outliers or influential cases via leave-one-out methods for data analysis. If a study was detected as an outlier or influential case, it was removed from the meta-analysis. Publication bias was assessed through the inspection of contour-enhanced funnel plots and Egger's bias tests (Egger et al., 1997). In instances of significant bias, Duval and Tweedie's trim-and-fill method was applied (Duval & Tweedie, 2000), alongside the calculation of the number of unpublished studies needed to negate the observed effect ($p > 0.05$).

To identify potential moderators affecting the efficacy of AE, random effects meta-regression analyses were performed via the maximum likelihood estimation method for model building. The dose−response relationship was evaluated through linear, quadratic, and cubic polynomial models, with the selection of the most optimal model based on the Akaike information criterion corrected (AICc) (Kletting & Glatting, 2009). Subgroup analyses were conducted to investigate the impact of several factors, such as the clinical diagnosis of depression at baseline, mean age, duration, intensity, type, and regimen length.

The "rma", "forest", "baujat", "fitstats" and "regplot" functions from the "metafor" package were employed to perform the meta-analysis (Viechtbauer, 2010). All analyses were conducted using R (R Foundation for Statistical Computing, Vienna, Austria).

3. Results

Upon searching, 1421 potentially relevant studies were identified across nine databases. Following the removal of duplicates, 782 potentially eligible studies remained for which abstracts were screened. After excluding studies on the basis of the full text, a total of 26 studies (comprising 22 different samples) were included in the main analysis. (Fig. 1).

3.1. Study characteristics

This meta-analysis involved 1308 participants, with 648 in the AE intervention group and 660 in the control group. Twenty-six studies focused on depressive symptoms, 18 of which reported a diagnosis of depression at baseline. The participant demographics were diverse, with female participation ranging from 26% to 100%. The age distributions encompass the period from childhood (9.3 years) to early adulthood (21.42 years). Specifically, the analysis encompassed studies involving children (n = 2) (Petty et al., 2009; Weintraub et al., 2008), early adolescence (n = 3) (Crews et al., 2004; Daley et al., 2006; Silva et al., 2020), late adolescence (n = 11) (Carter et al., 2015; COSTIGAN et al., 2016; Dabidi Roshan et al., 2011; Goldfield et al., 2015; He & Ji, 2004; Hughes et al., 2013; JEONG et al., 2005; Nabkasorn et al., 2006; Zhao et al., 2023; Zhu & Yu, 2011), and young adulthood (n = 10) (Hemat-Far et al., 2012; Jiao, 2010; Li, 2008; LI et al., 2014; Li et al., 2009; Sadeghi et al., 2016). The education stages included primary school (n = 4), high school (n = 7), and college (n = 15) (Table S1).

The regimen of the AE interventions exhibited considerable variability, ranging from 6 to 26 weeks. Typically conducted 1–5 times per week, with the highest frequency occurring 3 times per week (n = 18). The duration of AE lasted 8–85 min, featuring light to vigorous intensity (50–80% HRmax or 49–60% HRR). The types of AE included running (n = 10), dancing (n = 6), mixed sports (n = 6), high-intensity interval training (n = 1), swimming (n = 1), pool walking (n = 1), and football (n = 1). Eighteen studies implemented interval AE, whereas eight opted for continuous AE. Twenty-two studies highlighted team-based AE, with six detailing AE interventions conducted under supervision (Table S2). One study reported minor knee or ankle pain and delayed muscle soreness (Zhao et al., 2023), whereas another cited adverse events not related to the study (Weintraub et al., 2008).

The risk of bias assessment was conducted using the PEDro scale, and the methodological quality of the included studies was fair-to-good, as indicated by PEDro scale scores ranging from 5 to 8. Eighteen studies were rated as good quality (Carter et al., 2015; COSTIGAN et al., 2016; Crews et al., 2004; Dabidi Roshan et al., 2011; Daley et al., 2006; Goldfield et al., 2015; He & Ji, 2004; Hemat-Far et al., 2012; Hughes et al., 2013; JEONG et al., 2005; Jiao, 2010; LI et al., 2014; Li et al., 2009; Nabkasorn et al., 2006; Petty et al., 2009; Sadeghi et al., 2016; Weintraub et al., 2008; Zhao et al., 2023), and four studies were rated as fair quality (Chen & Diao, 2007; Li, 2008; Silva et al., 2020; Zhu & Yu, 2011). A summary of the methodological quality assessment for each included study and individual rating scales is provided in Table S3.

3.2. Main analysis

A meta-analysis of 26 studies revealed that AE notably decreased depressive symptoms among youth [$g = -0.92$; 95% CI (−1.16, −0.69); $p < 0.01$] (Fig. 2). There was a moderate degree of heterogeneity between studies ($\tau^2 = 0.26$; $I^2 = 74\%$), and the Q-test revealed a significant heterogeneity between studies (Q = 88.70; df = 25; $p < 0.01$). Publication bias was indicated by Egger's test (intercept = −2.66, $p < 0.05$). However, Duval and Tweedies' trim and fill method did not affect the effect. The failure-safe number of additional negative studies was 1272. The visual inspection of the funnel plot did not indicate a risk of bias (Fig. S1). Sensitivity analysis conducted using the leave-one-out method, demonstrated that the exclusion of individual studies did not

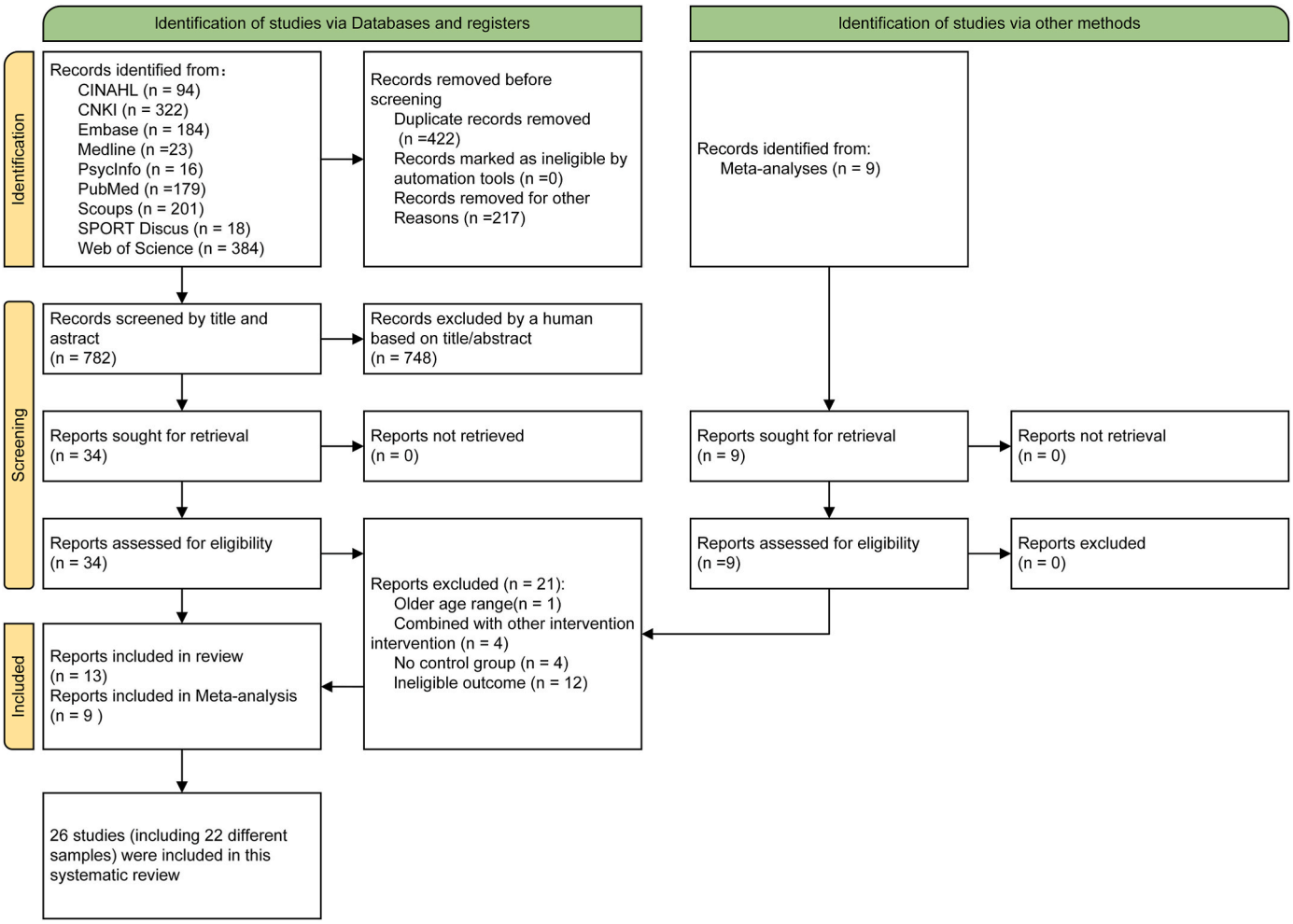

Fig. 1. PRISMA flow chart of the study selection.

substantially affect the outcome indicators. The pooled effect sizes remained within the original 95% CI after each study was excluded separately, suggesting the study results' robustness as shown in Table S4.

3.4. Subgroup analysis

3.4.1. Depression diagnosis

Among 16 studies involving participants with a depression diagnosis at baseline, AE had a significant overall effect, with a large effect size [$g = -1.13$; 95% CI (−1.40, −0.85); $p < 0.01$; $I^2 = 61.3\%$]. In 10 studies involving non-depressed individuals, AE also significantly decreased depression symptoms, with a smaller effect than the depressed subgroup [$g = -0.59$; 95% CI (−0.84, −0.34); $p < 0.01$; $I^2 = 56.7\%$] (Table 1).

3.4.2. Population

A meta-analysis of 10 studies of young adulthood revealed a significant efficacy of AE in alleviating depressive symptoms [$g = -1.04$; 95% CI (−1.43, −0.64); $p < 0.01$; $I^2 = 67.6\%$]. Similarly, 11 studies focusing on late adolescence reported a significant decrease [$g = -0.97$; 95% CI (−1.25, −0.69); $p < 0.01$; $I^2 = 61.3\%$]. In contrast, no significant effect of AE was observed in early adolescence or childhood ($p > 0.05$).

3.4.3. Education stage

AE showed the most substantial benefit among college students. An analysis of 15 studies indicated a significant reduction in depressive symptoms [$g = -1.04$; 95% CI (−1.32, −0.77); $p < 0.01$; $I^2 = 61.8\%$]. High school students also benefited, with an analysis of 7 studies showing a significant overall effect [$g = -0.81$; 95% CI (−1.20, −0.43); $p < 0.01$; $I^2 = 69.5\%$]. However, in primary school students, AE did not significantly reduce depressive symptoms, according to the combined results of 4 studies [$g = -0.90$; 95% CI (−2.24, 0.44); $p > 0.05$; $I^2 = 94.8\%$].

3.4.4. Duration

Analysis of 14 studies revealed that an AE duration between 25 and 40 min significantly reduced depressive symptoms [$g = -1.12$; 95% CI (−1.44, −0.80); $p < 0.01$; $I^2 = 74.8\%$]. Additionally, a longer duration (>40 min) in 8 studies also yielded significant improvements [$g = -0.94$; 95% CI (−1.33, −0.54); $p < 0.01$; $I^2 = 58.2\%$]. In contrast, an AE duration of less than 25 min had a smaller effect [$g = -0.26$; 95% CI (−0.50, −0.22); $p < 0.05$; $I^2 = 0\%$].

3.4.5. Intensity

One study reported that vigorous-intensity AE was significantly effective in diminishing depressive symptoms [$g = -1.57$; 95% CI (−2.05, −1.09); $p < 0.01$]. The analysis of 13 studies on moderate-intensity AE also revealed a significant reduction [$g = -1.05$; 95% CI (−1.38, −0.73); $p < 0.01$; $I^2 = 74.2\%$]. Moreover, the pooled results from 5 studies of light-intensity AE also indicated a significant beneficial effect [$g = -0.58$; 95% CI (−0.99, −0.18); $p < 0.01$; $I^2 = 51.7\%$].

3.4.6. Length of the AE regimen

Subgroup analysis revealed that AE regimens lasting longer than 16

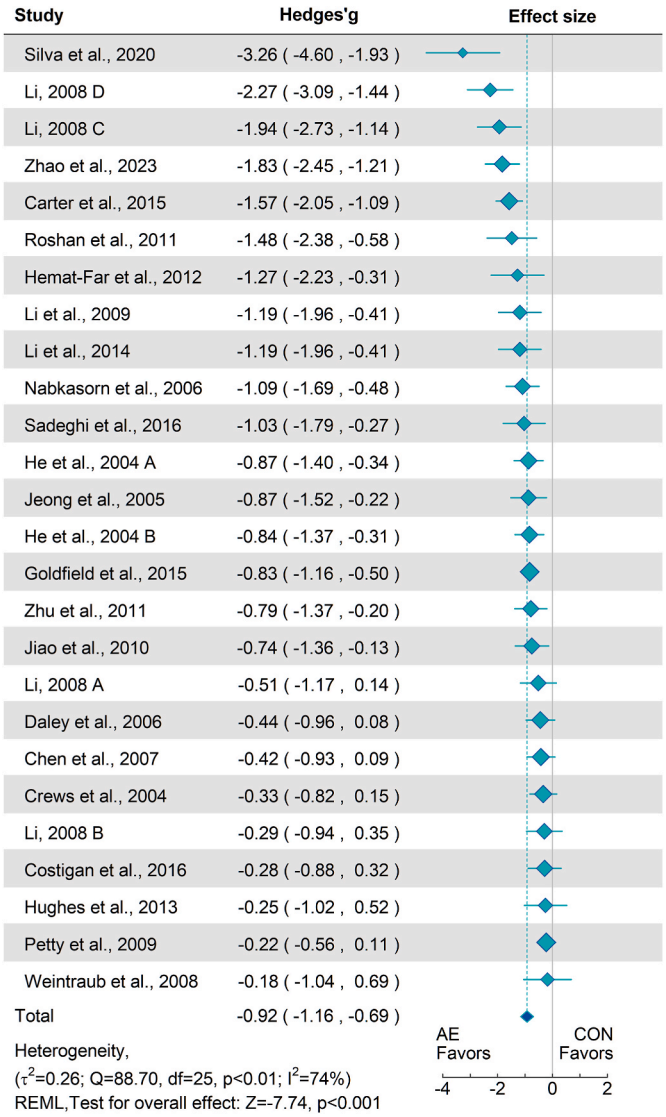

Fig. 2. Forest Plot of overall studies.

weeks had a significant beneficial effect on depression symptoms in 4 studies [$g = -0.79$; 95% CI (-1.03, -0.55); $p < 0.01$; $I^2 = 0\%$]. Similarly, eight studies examining AE durations of 9–15 weeks reported a notable reduction [$g = -0.99$; 95% CI (-1.58, -0.41); $p < 0.01$; $I^2 = 85.7\%$]. Pooled results from 14 studies with shorter regimens (<8 weeks) also revealed significant benefits [$g = -0.95$; 95% CI (-1.25, -0.65); $p < 0.01$; $I^2 = 65.6\%$].

3.4.7. Types of AE

Nine studies reported that running form-based AE significantly reduced depressive symptoms [$g = -1.02$; 95% CI (-1.23, -0.82); $p < 0.01$; $I^2 = 16.2\%$]. Similarly, dancing (n = 6) also reported a significant reduction [$g = -1.01$; 95% CI (-1.66, -0.36); $p < 0.01$; $I^2 = 82.7\%$]. Furthermore, a mixed form-based AE proved efficacious in alleviating depressive symptoms, as evidenced by six studies [$g = -0.60$; 95% CI (-1.03, -0.17); $p < 0.01$; $I^2 = 76.7\%$].

3.5. Effect of dose–response

In the meta-regression analysis, the AICc test revealed that the linear model achieved an optimal balance between goodness of fit and model simplicity except for the duration of the AE (Table 2). A significant dose–response relationship was observed between age and the alleviation of depression by AE [$\beta = -0.06$; 95% CI (-0.12, 0.00); $p < 0.05$]. Additionally, the intensity of AE was positively associated with reduced depression symptoms [$\beta = -0.03$; 95% CI (-0.06, 0.00); $p < 0.05$]. Considering the duration of AE, the AICc indicated that the quadratic model exhibited the optimal fit, with duration^2 evincing a positive correlation with depression symptom alleviation [$\beta = 0.001$; 95% CI (0.001, -0.002); $p < 0.05$], indicating that an AE duration of 25–40 min presented a notable advantage (Fig. 3).

4. Discussion

4.1. Main findings

To our knowledge, this research represents the inaugural systematic review examining the impact of aerobic exercise interventions on youth depression. The pooled results of the 26 studies included in the analysis suggest that AE significantly alleviates depressive symptoms in young people. Subgroup analysis revealed noteworthy positive effects of AE across several dimensions, and the analysis differentiated by education stage revealed that college students experienced greater benefits than high school students. Furthermore, moderate-intensity AE was more efficacious than light-intensity, adjusted analyses suggested that individuals diagnosed with mental illness or clinical depression derived more substantial benefits from the AE interventions than their non-depressed counterparts. The present study did not find team-based and supervision to be significant moderators of the effectiveness of AE interventions.

4.2. Effect of the AE training protocol

Our findings indicate that AE significantly influences depressive symptoms in late adolescence (high school) and early adulthood (college); however, no significant effect was observed in younger participants (under 14 years or in primary school). Notably, none of the studies in the primary school subgroup involved participants diagnosed with clinical depression at baseline (Crews et al., 2004; Petty et al., 2009; Silva et al., 2020; Weintraub et al., 2008). This implies that mental health disorders such as depression are less prevalent among those in early adolescence or younger, and that lower baseline outcomes may result in relatively slight changes. Furthermore, the observed dose–response relationship lends additional support to the robustness of the result, namely that the efficacy of AE increases with the participants' age. This finding aligns with a previous systematic review, which reported greater benefits of physical activity for participants aged 13 years or older than younger participants (Bailey et al., 2018). Differences in physical activity levels may account for the varying benefits observed across different age groups in response to AE interventions. A cohort study highlighted a decline in physical activity levels among adolescents aged 12–16 years, coupled with an increase in sedentary behavior, potentially worsening depressive symptoms (Kandola et al., 2020). This suggests that adolescents becoming more sedentary with age might respond more positively to AE interventions, whereas younger children, who are already more active, exhibit less responsiveness to additional physical activity stimuli.

Effect modification analyses revealed that the duration of AE moderated the effectiveness of AE in alleviating depressive symptoms, resulting in a significant quadratic dose–response relationship between the duration of AE and its impact on reducing depressive symptoms. Our findings revealed that 25–40 min of AE may lead to optimal psychological adjustments, and achieving the same benefit level may be more challenging with either shorter or longer durations, which provides valuable insights for clinical practice. This finding is supported by previous research, a systematic review encompassing all forms of exercise among youth suggested that 30 min can be beneficial (Li et al., 2023). Additionally, studies focused on adult depression have revealed a positive correlation between exercise duration and symptom improvement

Table 1
Subgroup and moderation meta-analysis.

	k	*g*	95% CI †	p_d ‡	I^2 (%) §	p_h ¶	p_f **	p_m ††
Depression Diagnosis								
Clinical depression	16	−1.13	−1.40, −0.85	<0.01	61.3	<0.01	0.76	0.01
Non-depressed	10	−0.59	−0.84, −0.34	<0.01	56.7	<0.01	0.23	
Population								
Young adulthood	10	−1.04	−1.43, −0.64	<0.01	67.6	<0.01	0.01	0.11
Late adolescent	11	−0.97	−1.25, −0.69	<0.01	61.3	<0.01	0.90	
Early adolescent	3	−1.25	−3.03, 0.53	0.17	85.4	<0.01	<0.01	
Childhood	2	−0.22	−0.53, 0.10	0.17	0	0.92	/	
Education stage								
Primary school	4	−0.90	−2.24, 0.44	0.19	94.8	<0.01	0.03	0.18
High school	7	−0.81	−1.20, −0.43	<0.01	69.5	<0.01	0.83	
College	15	−1.04	−1.32, −0.77	<0.01	61.8	<0.01	0.25	
Duration (min)								
>40	8	−0.94	−1.33, −0.54	<0.01	58.2	0.02	0.15	0.12
25–40	14	−1.12	−1.44, −0.80	<0.01	74.8	<0.01	0.08	
<25	4	−0.26	−0.50, −0.22	0.03	0	0.99	0.55	
Intensity								
Vigorous	1	−1.57	−2.05, −1.09	<0.01	/			0.94
Moderate	13	−1.05	−1.38, −0.73	<0.01	74.2	<0.01	0.05	
Light	5	−0.58	−0.99, −0.18	<0.01	51.7	0.08	0.07	
Length of intervention (Wek)								
≥16	4	−0.79	−1.03, −0.55	<0.01	0	0.55	0.26	0.54
9~15	8	−0.99	−1.58, −0.41	<0.01	85.7	<0.01	0.09	
≤8	14	−0.95	−1.25, −0.65	<0.01	65.6	<0.01	0.02	
Aerobic type								
Running	10	−1.02	−1.23, −0.82	<0.01	16.2	0.37	0.13	0.55
Dancing	6	−1.01	−1.66, −0.36	<0.01	82.7	<0.01	0.03	
Mixed	6	−0.60	−1.03, −0.17	<0.01	76.7	<0.01	0.68	

Note: †, 95% CI for pooled mean difference; ‡, *p* value for pooled mean difference; §, I^2 measure of heterogeneity between studies expressed as percentage; ¶, *p* value from the test of heterogeneity; **, *p* value for funnel plot asymmetry; ††, *p* value for the test of moderation effect.

Table 2
Dose−response effects of aerobic exercise on depression symptoms: meta regression coefficients from the AICc for the linear, quadratic, and cubic models.

	b_0 *	95% CI †	b_1 ‡	95% CI §	*p*‖	AICc ¶	$AICc_2$ **	$AICc_3$ ††
Main analysis								
Age	0.23	−0.76, 1.22	−0.06	−0.12, 0.00	0.03	52.88	55.32	58.00
Intensity	1.17	−0.96, 3.31	−0.03	−0.06, 0.00	0.04	34.41	37.70	41.42
Duration	0.82	−0.58, 2.22	−0.01	−0.17, −0.01	0.02	55.32	53.48	56.11
Duration ^ 2			0.001	0.001, 0.002	0.04			
Intervention period	−0.94	−1.58, −0.30	−0.09	−0.17, −0.01	0.78	59.31	60.90	64.15
Frequency	−0.86	−1.65, −0.06	0.00	−0.25, 0.25	0.99	59.37	62.11	64.93
Clinical depression only								
Age	1.05	−1.80, 3.89	−0.11	−0.26, 0.04	0.15	33.31	36.40	42.45
Intensity	−1.00	−1.47, 3.48	−0.03	−0.07, 0.01	0.07	25.47	32.85	51.25
Duration	−0.69	−1.69, 0.31	−0.01	−0.03, 0.01	0.38	32.73	36.12	41.03
Intervention period	−1.37	−2.58, −0.15	0.03	−0.10, 0.15	0.68	33.40	36.16	36.16
Frequency	−0.92	−1.88, 0.05	−0.07	−0.37, 0.23	0.65	33.36	36.23	40.81

Note: *, Intercept; †, 95% CI for b_0; ‡, Unstandardized regression coefficient; §, 95% CI for b_1; ‖, *p* value for b_1; ¶, Bias-corrected AICc for the linear model; **, Bias-corrected AICc for the quadratic polynomial model; ††, Bias-corrected AICc for the cubic polynomial model.

(Roeh et al., 2019). Previous studies examining exercise across diverse populations have indicated that a duration of 45–49 min produces superior outcomes compared to shorter and longer exercise durations (Rethorst et al., 2009). These findings lend support to our results, which indicated that, compared with a shorter duration, a longer AE duration may be more efficient. Excessive prolonged exercise, which activates proprotein synthesis and metabolic androgenic pathways, is also important to note, as it can lead to adverse emotional effects, such as increased irritability and aggression (Kersey, 1993). The World Health Organization's 2020 guidelines recommend that children and adolescents engage in 60 min of moderate-to-vigorous physical activity daily, primarily through AE, to promote overall health (Chaput et al., 2020). Hence, based on a comprehensive consideration of previous research and a focus on AE, the antidepressant efficacy of AE appears to be more pronounced when the duration ranges from 25 to 40 min.

The period of the AE regimen is also a crucial factor in determining its effectiveness in alleviating depressive symptoms. Our results indicated that the AE regimen spanning 9–15 weeks was particularly beneficial. This finding is partially supported by previous studies, which suggest that 12 weeks of exercise is generally optimal for achieving positive outcomes (Recchia et al., 2023). A detailed evaluation of diverse exercise programs revealed a marked decrease in depressive symptoms following 12 weeks of consistent physical activity (Li et al., 2023; Nasstasia et al., 2015). Given that physical and psychological adaptations to AE require a longer duration, this may explain why our study observed better results with a regimen lasting 9–15 weeks. Regular AE has also been associated with an increase in hippocampal gray matter volume, with research revealing that engaging in AE for more than 6 weeks can lead to increased volume of the hippocampus and cortical regions (Ferrer-Uris et al., 2022; Hendrikse et al., 2022). Notably, the hippocampus is one of the areas most affected in the brains of individuals with depression, experiencing an approximately 5% reduction in volume (Schmaal et al., 2016). Furthermore, a year-long AE regimen can result in a 2% increase in hippocampal volume; however,

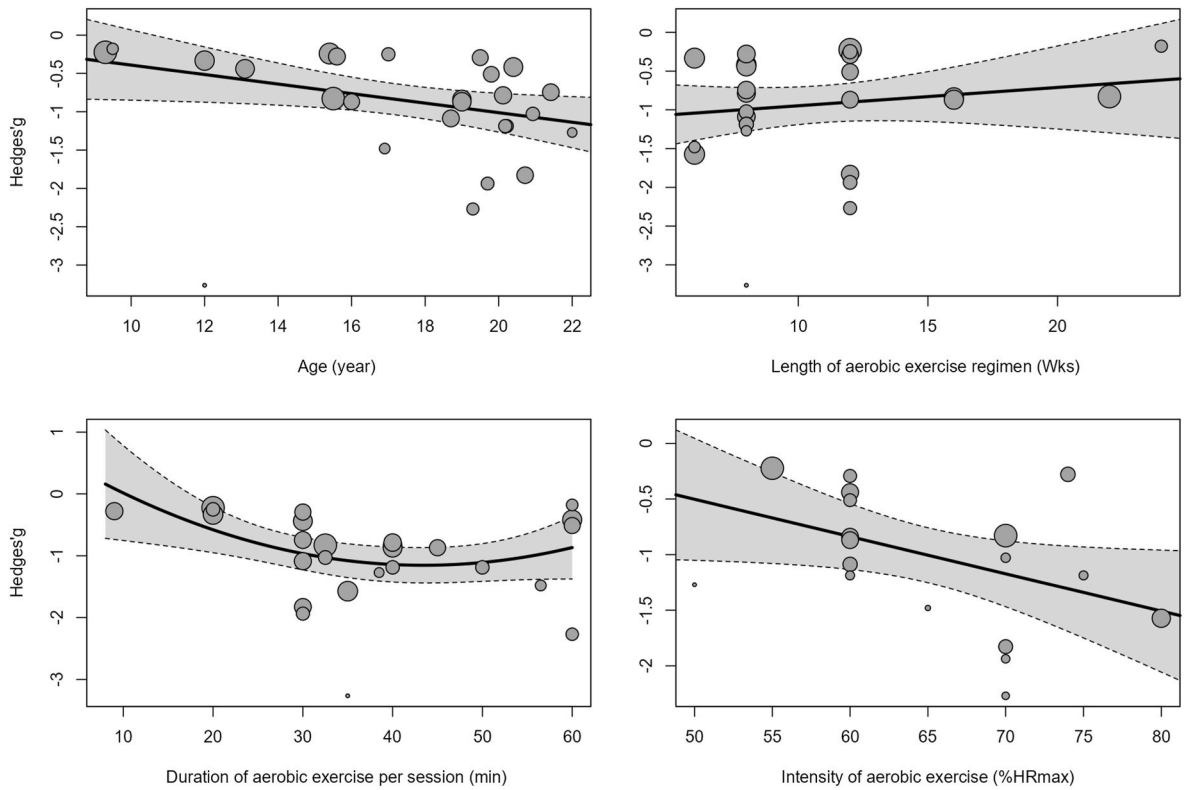

Fig. 3. Dose−response effect plot of aerobic exercise on depressive symptoms: results of meta- regression analysis for predictors.

discontinuation of AE leads to a return to baseline levels (Hendrikse et al., 2022; Xie et al., 2021). Thus, long-term AE plays a vital role in alleviating depressive symptoms by enhancing hippocampal volume. These findings support the idea that AE extending beyond 12 weeks may be as effective as pharmacotherapy in reducing depression (Blumenthal et al., 2007). Nonetheless, some studies suggest that extended intervention periods may have adverse effects, which raises the concern that simply increasing overall physical activity may not ensure sustained reductions in depressive symptoms (Kim et al., 2018; Pearce et al., 2022). Additionally, Bailey et al. highlighted the need to account for diminished attention spans in long-term interventions involving children and adolescents (Bailey et al., 2017). Therefore, our research recommends an intervention period of less than 15 weeks for optimal results.

The efficacy of AE in alleviating depressive symptoms is significantly related to the intensity of AE. While only one study was classified as involving vigorous-intensity in our analysis, subgroup analyses indicated that moderate-intensity AE ($g = -1.05$) had a more significant effect than light-intensity AE ($g = -0.58$). This finding is supported by Li et al., whose study focusing on young individuals assessed the impact of varying AE intensities on depressive symptoms and revealed that moderate-to-vigorousintensity AE was more efficacious in alleviating depression. Clinical evidence recommends that moderate-intensity exercise can promote various physiological adaptations that mitigate symptoms of depression (Anderson & Shivakumar, 2013). Hendrikse et al. reported that exercise could reverse structural brain damage induced by depression, with higher-intensity exercise resulting in greater benefits (Hendrikse et al., 2022). The pathophysiological mechanisms underlying depression involve alterations in the brain's neural systems, particularly a reduction in brain-derived neurotrophic factor (BDNF) activity. This reduction may lead to cortical dysfunction and neuronal impairment, subsequently facilitating the development of depression (Castrén & Rantamäki, 2010; Markham et al., 2014). Evidence suggests that BDNF levels appear to be altered by long-term AE (El-Sayes et al., 2019). Previous systematic reviews have indicated that moderate-to-vigorous intensity AE (65% VO2max), performed 2–3 times per week for at least 40 min per session, can effectively increase BDNF levels (Feter et al., 2019). The response of BDNF may correlate with AE intensity, Ferris et al. found that following higher-intensity AE (75% VO2max) compared to moderate-intensity AE (56% VO2max), BDNF levels increased, although this increase did not reach statistical significance (Ferris et al., 2007). Ross et al. compared vigorous-intensity and light-intensity AE, and demonstrated that only vigorous-intensity AE significantly elevated BDNF levels (Ross et al., 2019). These results align with our findings, suggesting that higher intensity of AE may yield larger effect sizes. A review focusing on adolescents posits that moderate-to-vigorous intensity AE appears to have positive effects on the developing brain, especially considering that adolescents and young individuals are typically at a developmental stage well below that of adults (Ferrer-Uris et al., 2022). This evidence suggests that when implementing AE as an antidepressant strategy, it is necessary to consider maintaining the intensity at moderate or higher.

Regarding the effect of the type of the AE regimen. Running form-based AE was identified as having the most substantial overall effect on reducing depressive symptoms ($g = -1.02$), closely followed by dance-based aerobic interventions ($g = -1.01$), albeit with significant heterogeneity among studies ($I^2 = 82.7\%$). Mixed exercise types demonstrated a lesser impact ($g = -0.60$). These findings align with those reported by Heissel et al. in a 2023 systematic review focusing on adults, suggesting that variations in the efficacy of different exercise types might stem from their intensity levels (Heissel et al., 2023). A 2015 Cochrane review reported that Dance Movement Therapy (DMT) at lower intensities did not significantly impact adolescent depression (Meekums et al., 2015). Conversely, aerobic dancing exhibited effects comparable to running form-based AE (Chen & Diao, 2007; He & Ji, 2004), with running showing notable significance and lower heterogeneity ($I^2 = 23.8\%$). Long-term running has been demonstrated to encourage synaptic formation, stabilize new synapses in cortical regions, and ameliorate anxiety-like behaviors due to activity restrictions (Chen et al., 2019). The inclusion of two water form-based AE in this

analysis resulted in significant positive outcomes. Recognized as a distinctive category of AE, water sports could enhance mood and alleviate anxiety, as supported by multiple studies (Silva et al., 2020). These activities trigger the release of β-endorphin and dopamine, yielding calming effects that significantly mitigate depressive symptoms. Moreover, the resistance offered by water, which exceeds that of land by more than tenfold, demands increased engagement from the motor cortex, thus providing more robust stimulation. The participants reported a unique sensation of water gliding over their skin, a tactile experience that reduces stress and promotes a sense of peace (Espí-López et al., 2020; von Mohr et al., 2017).

Dancing form-based AE studies have shown that dance markedly improves the well-being of youth, notably through emotional uplift and the diminution of self-reported depressive symptoms (Mansfield et al., 2018). Engagement in dance, whether as a participant or spectator, along with other performance-oriented sports activities, fosters positive well-being attributes, including excitement, purpose, increased confidence, self-esteem, and a sense of belonging (Badura et al., 2015). The potential connection between dance and mirror neurons has been explored to understand the underlying mechanisms involved. Mirror neurons are those that fire both when an individual performs an action and when they see someone else executing the same action or expressing an emotion, potentially leading the observer to mirror the positive emotions associated with the action (Gazzola et al., 2006). This empathy and active "mirroring" process is thought to be partly facilitated by the activation of these neurons (Winters, 2008). As an expressive art form, dance significantly influences emotional regulation through its intrinsic emotional components (Dieterich-Hartwell, 2017). The amalgamation of music and movement in dance can divert attention from adverse thoughts and experiences, substituting them with positive ones (Daviu et al., 2019).

Unfortunately, in our study, the frequency of AE did not emerge as a significant moderating factor. Most studies implemented interventions three times per week, although slightly more effective than once or five times a week, with no significant differences. Therefore, caution is warranted in interpreting these findings due to the limited number of studies that focused on AE interventions conducted once or five times a week. This finding is consistent with previous research, a cross-sectional study revealed a U-shaped relationship between physical activity frequency and mental health, indicating that engaging in exercise 10 to 15 times a month can yield optimal improvements in mental health (Xu et al., 2022). Furthermore, we found that team-based AE ($g = -0.89$) was more effective than individual AE ($g = -0.73$), although this difference also did not reach significance. Nevertheless, team-based AE undoubtedly provides a good platform for social interaction, which helps reduce feelings of loneliness and social isolation, enhances social support, and offers opportunities for young people with depression to communicate and share experiences with others (Delfin et al., 2022; Murray et al., 2021). Moreover, team exercise not only helps maintain interest and improve adherence to exercise regimens but also fosters positive emotional experiences and strengthens self-identity among participants (Boelens et al., 2021). Incorporating team sports into AE programs is particularly beneficial, as maintaining a higher exercise intensity can be challenging in individual workouts. This approach can enhance exercise effectiveness and boost participants' motivation and adherence through peer social interaction to relieve symptoms of depression (Feng et al., 2024). Additionally, the supervision of experienced professional coaches is important. In our study, supervised AE demonstrated greater effectiveness ($g = -0.92$), whereas unsupervised AE had lower effectiveness ($g = -0.38$). Supporting that supervision by professional coaches is essential for compliance with intervention protocols (Keller-Varady et al., 2016). For patients with MDD, supervised AE is particularly important because of their typically lower activity levels and higher sedentary behavior, which are often associated with negative symptoms such as a lack of motivation (Schuch et al., 2017; Vancampfort et al., 2013). Therefore, ensuring professional supervision in AE interventions can help regulate exercise intensity and training protocols, ultimately leading to more favorable outcomes.

In summary, the important findings of this systematic review could be used to guide the clinical treatment of depression. AE has been shown to be an effective intervention for alleviating depressive symptoms, particularly in individuals between late adolescence and early adulthood. Healthcare professionals may consider integrating team-based AE, such as running or aerobic dancing, for a period of 9–15 weeks with supervision by professional coaches into the treatment plan for patients with depression. This approach should be performed 3 times per week, with the intensity reaching a moderate to vigorous level ($\geq$40% HRR or $\geq$ 64% HRmax). Each AE duration should last 25–40 min, with adjustments made on the basis of the specific circumstances and preferences of the individual.

4.3. Limitations

This study has several limitations. The outcomes of the included meta-analyses were self-reported, which may have introduced bias compared with objective measurements made by a professional physician under their guidance. Furthermore, the lack of long-term follow-up data has constrained the generalizability of our results. Moreover, the description of exercise intensity is typically limited to the method of inclusion in the trial rather than being reported in the results. The absence of comprehensive monitoring of intensity, such as heart rate, during the intervention period, may introduce discrepancies between the expected and actual outcomes. The baseline physical activity levels of the participants were not reported in the studies reviewed, which could influence the observed effects of the aerobic interventions. For most AE protocol, our study found only the linear aspects of the dose–response relationship, addressing the direction but not the magnitude of the relationship between exercise and its effects. Future research should aim to include more trials to determine the optimal dosage of aerobic exercise for therapeutic efficacy. Despite the wide number of sensitivity analyses, meta-regressions, and subgroup analyses performed, the reasons behind the substantial heterogeneity found in our results could not be accurately identified. This issue should be considered when interpreting our meta-analyses.

4.4. Future research

An intriguing prospective agenda may encompass the enhancement and augmentation of the number of superior-quality clinical trials assessing the impact of these aerobic exercises on psychiatric disorders. Given the accessibility and generalizability of AE, as exemplified by the potential for integrating it into a physical education curriculum, a time-use epidemiological perspective on aerobic exercise to alleviate depressive symptoms in adolescents appears to be a promising avenue for further investigation. Furthermore, research into the potential mediating effects on the exercise–depression relationship is highly important to gain a deeper understanding of the specific mechanisms by which aerobic exercise alleviates depressive symptoms.

5. Conclusion

This systematic review aims to synthesize the evidence from current studies on the efficacy of aerobic exercise in alleviating depressive symptoms in adolescents. Our findings of 26 studies involving a total of 1308 participants and 22 distinct samples, suggest that aerobic exercise interventions can effectively alleviate depressive symptoms in youth. Our results support the use of aerobic exercise as a potential treatment for depression, despite the observed lower efficacy in younger adolescents. However, given the ease of integrating aerobic exercise into existing sports curricula, it still demonstrates promise as a preventive measure.

CRediT authorship contribution statement

Qiushi Yang: Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Junli Wang:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Yiyaochen Lu:** Writing – original draft, Methodology, Formal analysis, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.psychsport.2025.102803.

Data availability

Data will be made available on request.

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